

WORKING WITH TIME AND TEMPERATURE

You are already familiar with most units of **time**. You can make conversions with units of time using the same methods you use with other units. The most common units of time are shown in the table.

When you work with time, sometimes you will get an answer with a decimal. That decimal represents a part of a larger unit.

Unit	Equivalencies
second (s or sec)	1 s = $\frac{1}{60}$ min ✓
minute (min)	1 min = 60 s ✓
hour (hr)	1 hr = 60 min ✓
day (d)	1 d = 24 hr ✓
week (wk)	1 wk = 7 d ✓
month (mo)	1 mo = $\frac{1}{12}$ yr ✓
year (yr)	1 yr = 12 mo ✓ 1 yr = 365 d

Time Examples

Friday, October 11, 2024 11:28 AM

$$6 \text{ hrs} \div 4 \text{ clients} = 1\frac{1}{2} \text{ hrs each}$$

scratch
 $\frac{6}{4} = \frac{3}{2}$

WORKING WITH TIME

Example From 1:30 p.m. to 7:30 p.m., Olivia can meet with 4 clients. If she sees each client for an equal amount of time, how long does each meeting take? Express your answer in hours and minutes.

Step 1

Subtract.

$$\begin{array}{r} 7:30 \\ - 1:30 \\ \hline 6:00 = 6 \text{ hr } 0 \text{ min} \end{array}$$

Divide

$$6 \div 4 = 1.5 \text{ hours}$$

Step 2

Change to hours and minutes.

The 1 represents 1 whole hour.
The .5 represents a part of an hour.

$$1 \text{ hour} = 60 \text{ minutes}$$

$$.5 \times 60 = 30 \text{ minutes}$$

TIP

Be careful when calculating time with a calculator! 3.25 hours equals 3 hours 15 minutes, not 3 hours 25 minutes.

Your knowledge of fractions can also help you make the conversion. You know that .25 is equal to $\frac{1}{4}$. What is $\frac{1}{4}$ of 1 hour? 15 minutes

Olivia meets with each client for **1 hour and 30 minutes.**

Solve the following problems.

1. The running time of a movie is 102 minutes. What is the running time in hours and minutes?

$$\begin{aligned} 1 \text{ hr} &= 60 \text{ min} \\ 102 \text{ min} &= 1 \text{ hr } 42 \text{ min} \end{aligned}$$

3. Noelle's time card shows that she worked from 8:30 a.m. to 12:00 p.m. and from 1:35 p.m. to 5:25 p.m. How many hours and minutes did she work?
- $$\begin{aligned} 8:30 - 12:00 &= 3\frac{1}{2} \text{ hrs} \\ 1:35 - 5:25 &= 10 \text{ min less than } 4 \text{ hrs} \\ \therefore 7\frac{1}{2} \text{ hrs} - 10 \text{ min} &= 7 \text{ hrs } 20 \text{ min} \end{aligned}$$

2. A runner completes her first lap around a track in 2 minutes 58 seconds. She completes her second lap in 3 minutes 7 seconds. How many minutes and seconds does it take for her to run both laps?

$$\begin{aligned} 2 \text{ } \boxed{58} + 3 \text{ } \boxed{7} & \text{ Note: } \\ & \text{ ' means feet or minutes} \\ & \text{ '' means inches or seconds} \\ = 5 \text{ } \boxed{65} & \\ = 6 \text{ } \boxed{5} & \\ \therefore 6 \text{ min } 5 \text{ sec} & \end{aligned}$$

4. Reggie's car payment is \$284 per month. How much will he pay in 5 years for his car?

$$\begin{aligned} & \text{have paid } \{ \text{wrong tense} \\ 5 \text{ yrs } (12 \text{ months/yr}) &= 60 \\ 60 \text{ payments } (284 \text{ per payment}) & \\ = \$17,040 & \end{aligned}$$

Temperature

Monday, October 14, 2024 5:38 PM

technically, it only measures heat.
E.g., 0°K = the complete absence of heat.

Temperature measures the hotness or coldness of an object or environment. The two most common scales for measuring temperature are Fahrenheit ($^{\circ}\text{F}$) and Celsius ($^{\circ}\text{C}$). & Kelvin ($^{\circ}\text{K}$)

When you work with temperature, you will often have to decide whether a temperature is increasing or decreasing. An increase in temperature is sometimes written with a + sign, and a decrease is sometimes written with a - sign.

warning: $+5^{\circ}\text{F}$ is 5 degrees Fahrenheit
 $+5^{\circ}$ means an increase of 5 degrees -5° means a decrease of 5 degrees
no body uses $+ -$ like this !!

WORKING WITH TEMPERATURE

Example The normal temperature for the human body is 98.6°F . Kyle has a temperature of 101.4°F . How many degrees higher than normal is Kyle's temperature?

Use subtraction to find the difference between two temperatures.

$$101.4^{\circ} - 98.6^{\circ} = 2.8^{\circ}\text{F}$$

Kyle's temperature is 2.8°F higher than normal, or $+2.8^{\circ}$.

$$\begin{array}{r} 101.4 \\ - 98.6 \\ \hline 2.8^{\circ}\text{F} \end{array}$$

also note no one ever says "I have a temperature 2.8°F above normal," people say, "I have a temperature of 101.4°F ." "I have a temperature" means "I have a temperature above normal."

Temperature Examples

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Solve the following problems.

5. The average August temperature in San Diego is 74°F . On August 4, it was 82°F . How far above or below the average was the temperature on August 4?

$$82 - 74 = 8$$

$\therefore 8^{\circ}\text{F}$ above the average

6. On a Celsius scale, the normal human body temperature is 37° . If Jae's temperature is 2.4°C higher than normal, what is his current temperature?

$$37 + 2.4 =$$

$$39.4$$

- A. 34.6°C
B. 37°C
C. 39.4°C
D. 41°C

7. During the evening, the temperature in Bellevue dropped 8°F . If the temperature was 67°F , what did the temperature drop to?

$$67 - 8 = 59$$

$\therefore 59^{\circ}\text{F}$

8. Kalia flies from Houston to Chicago. The temperature in Houston is 25.6°C . The temperature in Chicago is 3.3°C . Which amount represents the increase or decrease in temperature for Kalia?

$$25.6 - 3.3 =$$

$$22.3$$

- A. -28.9°C
B. -22.3°C
C. $+22.3^{\circ}\text{C}$
D. $+28.9^{\circ}\text{C}$